

This session, as the first one, aims at taking in hand some mathematical tools available to describe random elements. A set of various exercises are proposed. An numerical study will be carried out using Matlab to realize the information contained in the different mathematical entities, and their respective limits.

1 Cumulative distribution function and probability density function

1. Create a function in **Matlab** that reads a data set and plots the cumulative distribution function (CDF) from an experimental set of discrete data. Compare this function with the Matlab functions **ecdf**.
2. Estimate the probability density function (PDF) from an experimental set of data.
3. Estimate the cumulative distribution function from the probability density function.

2 Covariance and correlation

The covariance of two random variables is calculated by:

$$\text{Cov}(X, Y) = \mathbb{E}\{(X - \mathbb{E}\{X\})(Y - \mathbb{E}\{Y\})\}. \quad (1)$$

4. Implement a function which computes the covariance of two variables. Compare with the function **cov**.
5. Implement a function which computes the correlation between two variables. Compare with **corrcoef**.

Consider the random variables:

- r : the discrete random variable describing the number of cases,
 - u : the continuous random variable describing the mean mass of the cases with a uniform distribution,
 - n : the continuous random variable describing the mean mass of the cases with a normal distribution,
6. Using the function previously implemented or the Matlab function **cov**, investigate the covariance:
 - between the different random variables as previously defined,
 - between one of this random variable and itself,

Consider a random variable x uniformly distributed on $[-10; 10]$ and $y = x^2$, where y is completely determined by x , therefore x and y are not independent, but their correlation - as their covariance - is 0.

7. Which properties exhibits this matrix?
8. Using the function previously implemented or the Matlab function **corrcoef**, investigate the correlations between the different random variables (r, u, n) .

3 Random vectors

9. Simulate a random vector (select dimension greater than 1 in **getRandom.m**):
 - Choose a sample space and a probability distribution of the random experiment.
 - Simulate $n_1 = 10$, and $n_2 = 100$ numerical experiments of the vector.
 - Estimate the expected value and the covariance matrix of the random vector. How is this covariance matrix computed? (Hint: Use the **reshape** function to reorder the matrix containing the random variable in vectors). Compare results with the Matlab built-in function **cov(n, 2*n, 1)**.

4 Random fields

10. Plot one realization of a random field, e.g. along the length of a beam $x_0 = [0, 5]$ m, and the random variable relative to one position x_0 for:
 - a one-variate one-dimensional random field,
 - a two-variate one-dimensional random field.
 - a one-variate two-dimensional random field.
11. Choose some values of the random field, and estimate the values of the autocorrelation function of this random field for the relative points.
12. Investigate the autocorrelation function $C_{HH}(x_1, x_2) = \exp -\frac{|x_1 - x_2|}{b}$. How does the function evolve with respect to the parameter b ? What is the name of this parameter?

5 Two random variables

Consider the bivariate Gaussian probability distribution for Young's modulus E and Poisson's ratio ν . They have respectively the mean values μ_E and μ_ν , and the standard deviations σ_E and σ_ν .

The bivariate Gaussian probability density function $f_{E\nu}$ reads:

$$f_{E\nu}(E, \nu) = \frac{1}{2\pi\sigma_E\sigma_\nu\sqrt{1-\rho_{E\nu}^2}} \exp -\frac{1}{2(1-\rho_{E\nu}^2)} \left[\frac{(E-\mu_E)^2}{\sigma_E^2} + \frac{(\nu-\mu_\nu)^2}{\sigma_\nu^2} - \frac{2\rho_{E\nu}(E-\mu_E)(\nu-\mu_\nu)}{\sigma_E\sigma_\nu} \right] \quad (2)$$

where $\rho_{E\nu}$ is the correlation between the two random variables E and ν .

We consider the numerical application: $\mu_E = 21\,000$ MPa, $\sigma_E = 1050$ MPa, $\mu_\nu = 0.3$, $\sigma_\nu = 0.03$ and $\rho_{E\nu} = 0.20$.

13. Represent the joint probability density function $f_{E\nu}$. Does this function satisfy the condition:

$$\int_{\omega_X\omega_Y} f_{XY} dx dy = 1?$$

14. Represent the marginal probability density function of ν . Estimate the probability that $\nu = 0.25$.
15. Represent the marginal probability density function of E . Estimate the probability that $E = 21000$ MPa.
16. Estimate the probability that $E = 21000$ MPa while $\nu = 0.25$.
17. Estimate the probability that $E \leq 21000$ MPa while $\nu \leq 0.25$.
18. Are the two random variables E and ν independent?
19. Plot the conditional probability of E , given $\nu = \bar{\nu}$. Compare the probability density function of E to the probability density function of E given $\nu = 0.3$, for $\rho_{E\nu} = 0.20$, $\rho_{E\nu} = 0.40$ and $\rho_{E\nu} = 0.00$.

$$\int_{-\infty}^{\infty} \exp -\frac{(t-a)^2}{2\sigma^2} dt = \sqrt{2\pi}\sigma$$

Solution

List of Matlab files:

Exercise 1	getCdfPdf.m
Exercise 2	getCovarianceCorrelation.m
Exercise 3	getRandom.m getCovarianceCorrelation.m
Exercise 4	Rand_field.m correlation_law.m
Exercise 5	two_variables.m

Difficult points:

- Random fields: weakly homogeneous, strongly homogeneous and isotropic.

1 Cumulative distribution function and probability density function

1. Estimate the cumulative distribution function from an experimental set of discrete data. Compare this function with the Matlab functions `ecdf`.

Matlab file: `getCdfPdf.m`

```
A = 15 + 10*rand(1,500000);

x_min = 15;
x_max = 25;

nb_interval = 10000;
dx = (x_max - x_min) / nb_interval;
x = [x_min:dx:x_max];

B = sort(A);
j=1;

% Compute the cumulative distribution function
cdf = zeros(size(x));
for i = 1:length(x)
    j=1;
    if (x(i)-B(j)) > 0
        cdf(i) = j/length(B);
        for k = (j+1):length(B)
            if (x(i)-B(k)) > 0
                cdf(i) = k/length(B);
            end
        end
    else
        cdf(i) = 0;
    end
end
```

Alternatively, the cumulative distribution function can be calculated using logical indexing in Matlab:

```
cdf = zeros(size(x));
for i = 1:length(x)
    cdf(i) = numel(B(B<x(i)))/length(B);
end
```

Matlab function to get the cumulative distribution function:

```
[Fx,x] = ecdf(X)
```

2. Estimate the probability density function from an experimental set of data.

```
% Compute the probability density function
pdf = zeros(size(x));
for i = 1:length(x)
    if i==1
        pdf(i) = (cdf(i+1)-cdf(i))/(x(i+1)-x(i));
    elseif i== length(x)
        pdf(i) = (cdf(i)-cdf(i-1))/(x(i)-x(i-1));
    else
        pdf(i) = (cdf(i+1)-cdf(i-1))/(x(i+1)-x(i-1));
    end
end
```

3. Estimate the cumulative distribution function from the probability density function.

```
% Compute the cumulative distribution function from the proabability density function
cdf2(1)=0;
cdf2(length(x))=1;
for i= 2:length(x)-1
    cdf2(i)=cdf2(i-1)+ (pdf(i-1)+pdf(i+1))/2*(x(i)-x(i-1));
end
```

Example of plots:

2 Covariance and Correlation

4. Implement a function which computes the covariance of two variables. Compare with the Matlab function cov.

Matlab file: getCovarianceCorrelation.m

$$\text{Cov}(X, Y) = \mathbb{E}\{(X - \mathbb{E}\{X\})(Y - \mathbb{E}\{Y\})\}. \quad (3)$$

```
my_covar = zeros(2,2);
my_covar(1,1) = mean([A-mean(A)*ones(size(A))].*(A-mean(A)*ones(size(A))));
my_covar(1,2) = mean([A-mean(A)*ones(size(A))].*(B-mean(B)*ones(size(B))));
my_covar(2,1) = mean([B-mean(B)*ones(size(B))].*(A-mean(A)*ones(size(A))));
my_covar(2,2) = mean([B-mean(B)*ones(size(B))].*(B-mean(B)*ones(size(B))));
```

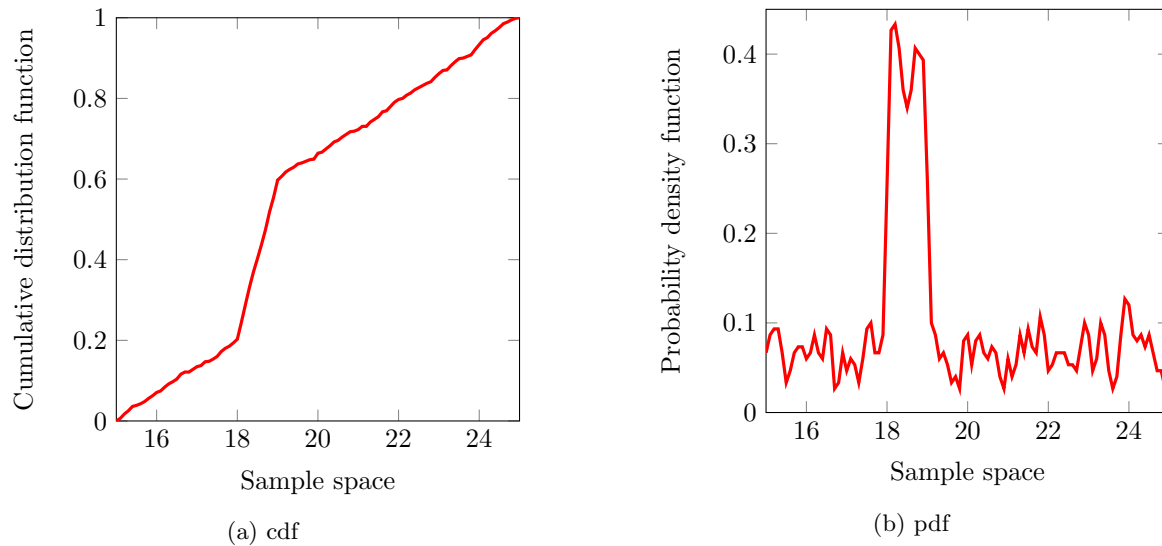


Figure 1: Cumulative distribution function and probability density function computed from a set of discrete data

- First, we consider $A = [1; 2]$, and $B = [1; 2]$. The covariance is equal to 0.25 when we compute with our own-implemented function (Eq.3). With the **Matlab** function `cov`:

```
cov(A,B)
```

we get 0.5. We get 0.25 when we use `cov(A,B,1)`. Why? By default, the **Matlab** covariance function is normalized to be unbiased: we get also 0.5 by multiplying Eq.3 by $N/(N-1)$:

```
my_covariance(A,B).*length(A)/(length(A)-1).
```

- Now, consider $A = [1; 2; 3]$, and $B = [1; 2; 3]$. The covariance is equal to $2/3$ by the formula (3) or the **Matlab** function `cov(A,B,1)`, and it is equal to 1 using the **Matlab** function `cov`.

- Now, consider $A = [1; 2; 3]$, and $B = [30; 20; 1]$. Using Eq.3 or `cov(A,B,1)`, the covariance matrix is equal to

$$\begin{bmatrix} 0.66667 & -9.666667 \\ -9.66667 & 144.666667 \end{bmatrix}.$$

With `cov(A,B)` with **Matlab**, we get

$$\begin{bmatrix} 1.000000 & -14.5000 \\ -14.5000 & 217.0000 \end{bmatrix}.$$

- Implement a function which computes the correlation between two variables. Compare with the **Matlab** function `corrcoef`.

```
my_corre = my_covar./(var(A,1)^0.5*var(B,1)^0.5);
```

This provides the same results as the Matlab function `corrcoef`. Note that the *matrix of correlation coefficients* always has diagonal 1 (as obtained with `corrcoef`), and only the coefficients outside the diagonal are of interest if the correlation coefficient is obtained as above. Alternatively, one can get the correlation coefficient as:

```
my_corre = zeros(2,2);
my_corre(1,1) = mean([A-mean(A)*ones(size(A))].*(A-mean(A)*ones(size(A))))/sqrt(var(A,1)*var(A,1));
my_corre(1,2) = mean([A-mean(A)*ones(size(A))].*(B-mean(B)*ones(size(B))))/sqrt(var(A,1)*var(B,1));
my_corre(2,1) = mean([B-mean(B)*ones(size(B))].*(A-mean(A)*ones(size(A))))/sqrt(var(B,1)*var(A,1));
my_corre(2,2) = mean([B-mean(B)*ones(size(B))].*(B-mean(B)*ones(size(B))))/sqrt(var(B,1)*var(B,1));
```

6. Using the function previously implemented or the Matlab function `cov`, investigate the covariance:

- between the different random variables as previously defined,
- between one of this random variable and itself,
- between the random masses and the random weights,
- between a random variable RV1 and another random variable defined from RV1.

7. Which properties exhibits this matrix?

The matrix is symmetric, and the diagonal contains the variance of each random component:

$$\text{Cov}(X, X) = \text{Var}(X) \quad (4)$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)^T \quad (5)$$

We can also see that:

$$\text{Cov}(cX, Y) = c * \text{Cov}(X, Y) \quad (6)$$

We can see that in the components (1,2) and (2,1) of the covariance matrix.

$$\text{Cov}(X + c, Y) = \text{Cov}(X, Y) \quad (7)$$

```
cov(u+15*ones(size(u)),r,1)
cov(u,r,1)
```

Algebraic formula for the variance - the König-Huygens formula:

$$\text{Cov}(X, Y) = \mathbb{E}\{X.Y^T\} - \mathbb{E}\{X\}.\mathbb{E}\{Y^T\} \quad (8)$$

For two vectors `u` and `r`, the components (1,2) and (2,1) of the covariance matrix:

```
cov(u,r,1)
```

are the same as:

```
mean(u.*r)-mean(u)*mean(r)
```

The matrix is positive semi-definite. All its eigenvalues are either positive or equal to 0:

```
eig(cov(u,r,1))
```

The diagonalization of this matrix allows a principal component analysis. Every positive definite matrix is invertible.

8. Using the function previously implemented or the Matlab function `corrcoef`, investigate the correlations between the different random variables.

```
corrcoef(w,2*w) =
ans =
    1    1
    1    1

corrcoef(w,-30*w)
ans =
    1.0000   -1.0000
   -1.0000    1.0000

corrcoef(u,2*w)=corrcoef(u,200*w)
```

The correlation is comprised between -1 and +1. The correlation - as the covariance - only informs about linear dependence. Let consider a random variable x uniformly distributed on $[-10;10]$ and $y = x^2$. y is completely determined by x , therefore x and y are not independent, but their correlation - as their covariance - is 0.

```
A = [-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10];
B = [100 81 64 49 36 25 16 9 4 1 0 1 4 9 16 25 36 49 64 81 100];

[g,kk] = my_covariance(A,B)

g = 1.0e+03 *
    0.0367   -0.0000
   -0.0000    1.0682

kk = 0.1853   -0.0000
   -0.0000    5.3975
```

Correlation could be more investigated. We could represent the set of data, the linear regression, and the correlation coefficient (see the Wikipedia page).

3 Random vectors

9. Simulate a random vector.

- Choose a sample space and a probability distribution of the random experiment:

Matlab file: `getRandom.m`

For example the force which is imposed to a beam: F_X with values between 100 and 300, F_Y with values between 400 and 500, and F_Z between 450 and 500.

```
% vector dimension
dim = 3;
```

```
% number of samples
samp = 100;

% Discrete RV, integer values between 100 and 150
r = randi([100 150],dim,samp);

% Continuous RV,
% uniformly distributed between 20 and 30
u = 20 + 10*rand(dim,samp);

% normal distribution
n = normrnd(25,5,[dim samp]);

% Weibull distribution
w = 20 + 10* wblrnd(0.5,0.4,[dim samp]);

m1 = r*30;
```

- Simulate $n_1 = 10$, and $n_2 = 100$ numerical experiments of the vector:
- Estimate the expected value and the covariance matrix of the random vector. How is this covariance matrix computed?

```
F(:,1) = normrnd(25,5,[dim samp]);
F(:,2) = normrnd(25,5,[dim samp]);
F(:,3) = normrnd(25,5,[dim samp]);
```

```
F_mean = mean(F);
```

The covariance matrix of the two matrices is computed considering the two vectors built from the matrices:

Matlab file: `getCovarianceCorrelation.m`

```
if size(A) ~= 1
    A = reshape(A',1,size(A,2)*size(A,1));
    B = reshape(B',1,size(B,2)*size(B,1));
end
```

Our own-implemented function gives an equivalent result as the function `cov(A,B,1)` when A and B are two matrices.

4 Random fields

10. Simulate a random field. Provide one realization of the random field, and the random variable relative to one position x_0 for:

- a n -dimensional field:

For example, for a beam $[0;5]$, we provide a random value: this is a one-dimensional uni-variate random field, uniformly distributed on Figure 2, and with a normal distribution on Figure 3.

Matlab file: `Rand.field.m`

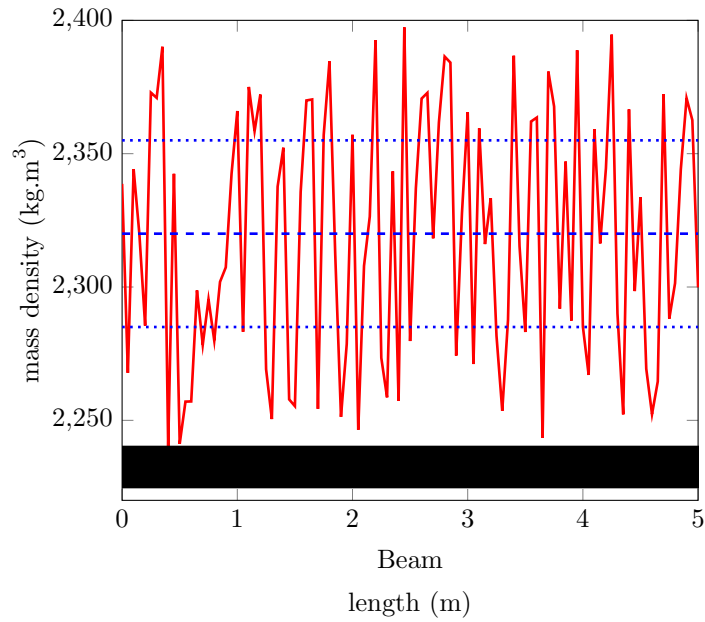


Figure 2: One realization of the univariate uni-dimensional random field representing a uniformly distributed between 2 240 kg.m³ and 2 400 kg.m³ random mass density - mean = 2 320 kg.m³ and standard deviation = 46 kg.m³

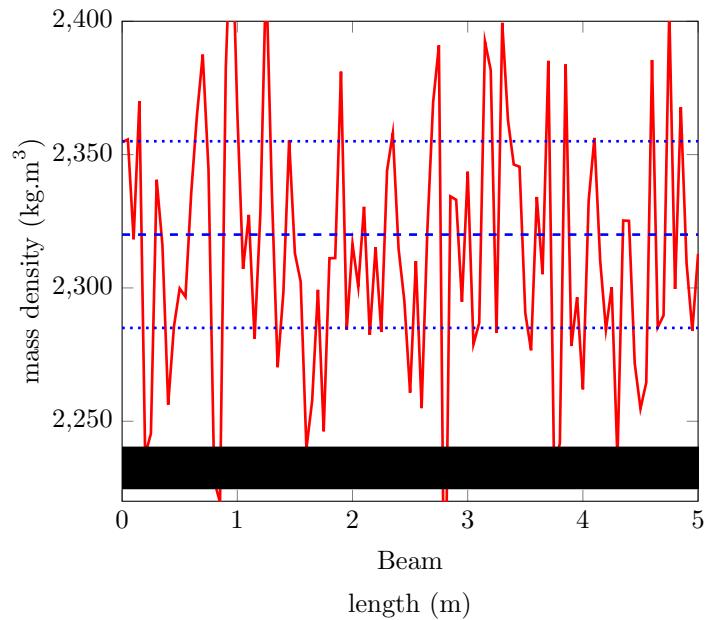


Figure 3: One realization of the univariate uni-dimensional random field representing a Gaussian distributed random mass density with mean = 2 320 kg.m³ and standard deviation = 46 kg.m³

For a plane $[0; 5]^2$, we provide a random value: this is a two-dimensional uni-variate random field (See Figure 4).

Matlab file: `Rand_field.m`

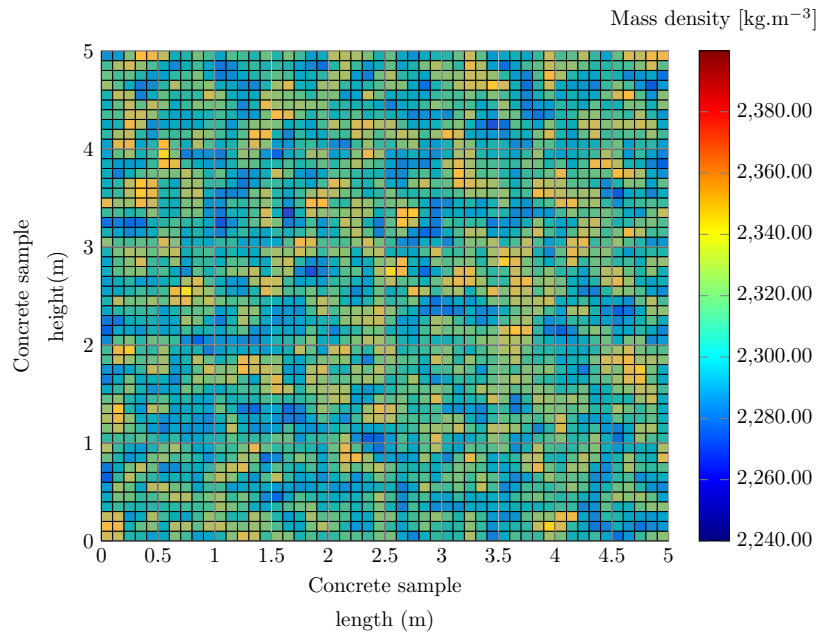


Figure 4: One realization of the univariate two-dimensional random field representing a uniformly distributed between 2 240 kg.m³ and 2 400 kg.m³ random mass density - mean = 2 320 kg.m³ and standard deviation = 46 kg.m³

For example, for a cube $[0; 1]^3$, we provide a random value: this is a three-dimensional uni-variate random field.

Matlab file: `Rand_field.m`

- a n -multivariate field:

for example, for a line $[0; 1]$, we provide two random values, one associated with u_x , the other one with u_y . This is a one-dimensional two-variate random field.

Matlab file: `Rand_field.m`

11. Choose some values of the random field, and estimate the values of the autocorrelation function of this random field for the relative points.

To estimate the autocorrelation of the random field, we need several realization of the random field.

Matlab file: `Rand_field.m`

```
% define the reference point
```

```
x0 = 2.5 ;
```

```
% estimate the index relative to the reference point
```

```
ind_x0 = (x0-x(1))/(x(2)-x(1)) +1;
```

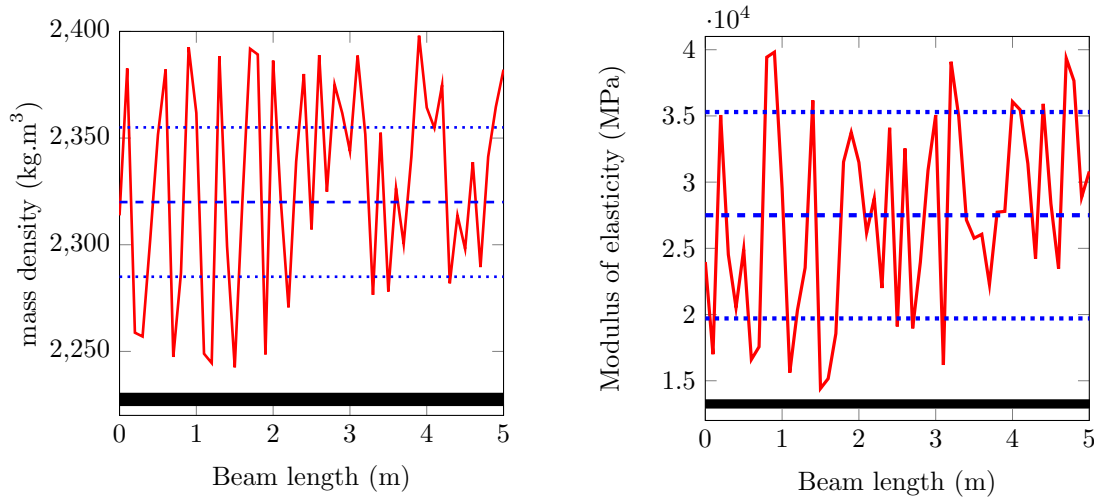


Figure 5: One realization of the two-variate uni-dimensional random field representing a uniformly distributed mass density with mean = 2 320 kg.m³ and standard deviation = 46 kg.m³ and a uniformly distributed modulus of elasticity with mean = 27 500 MPa and standard deviation = 7 794 MPa

```
autocor = zeros(1,length(x));
for kk = 1 : length(x)
    Acor = corrcoef(field(kk,1,:), field(ind_x0,1,:));
    autocor(1,kk) = Acor(1,2);
end

h2 = figure(2);
plot(x,autocor,'Linewidth',1.20)

xlim([0 5])
ylim([0 1.1])

xlabel({'Concrete sample','length (m)'},'Interpreter','Latex','FontSize',20)
ylabel('Correlation between x and the reference point x$_{0}$ = 2.5 m',
'Interpreter','Latex','FontSize',20)

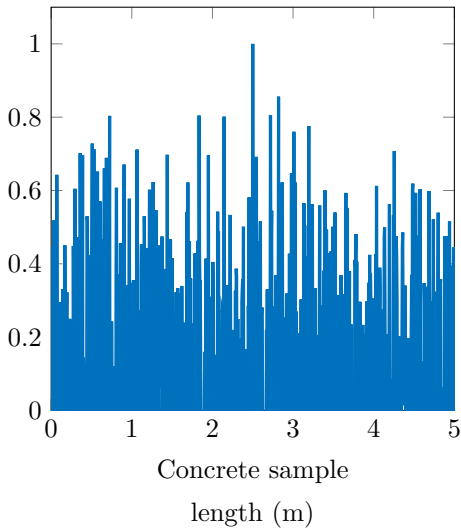
matlab2tikz('experimental_correlation.tikz')
```

12. Investigate the autocorrelation function $C_{HH}(x_1, x_2) = \exp -\frac{|x_1 - x_2|}{b}$. How does the function evolve with respect to the parameter b ? What is the name of this parameter?

Matlab file: `correlation_law.m`

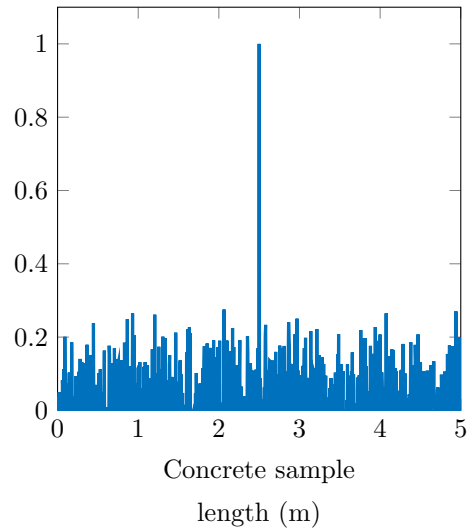
The parameter b is called the correlation length. Larger is the parameter b , more important are the long-range interaction. When the value of b is small, correlations are localized, and it appears no long-range interactions, the value of the correlation is almost equal to 0 far away from the reference point.

Correlation between x and the reference point $x_0 = 2.5$ m



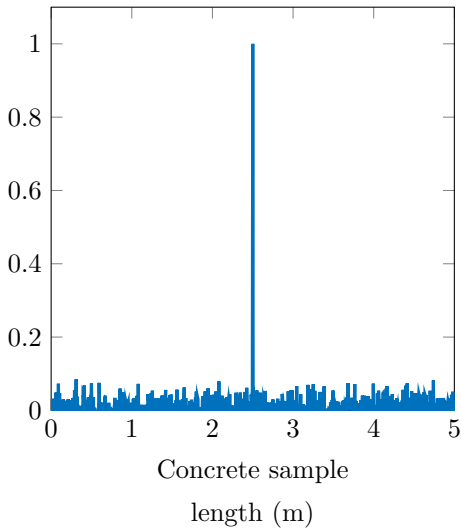
(a) 10 realizations of the random field

Correlation between x and the reference point $x_0 = 2.5$ m



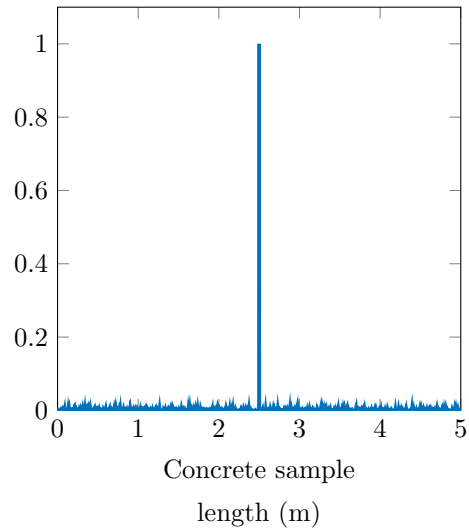
(b) 100 realizations of the random field

Correlation between x and the reference point $x_0 = 2.5$ m



(c) 1000 realizations of the random field

Correlation between x and the reference point $x_0 = 2.5$ m



(d) 100 000 realizations of the random field

Figure 6: Evolution of the correlation over the beam with respect to the reference point $x_0 = 2.5$ m considering different numbers of realizations of the random field and a uniform distribution (point discretization)

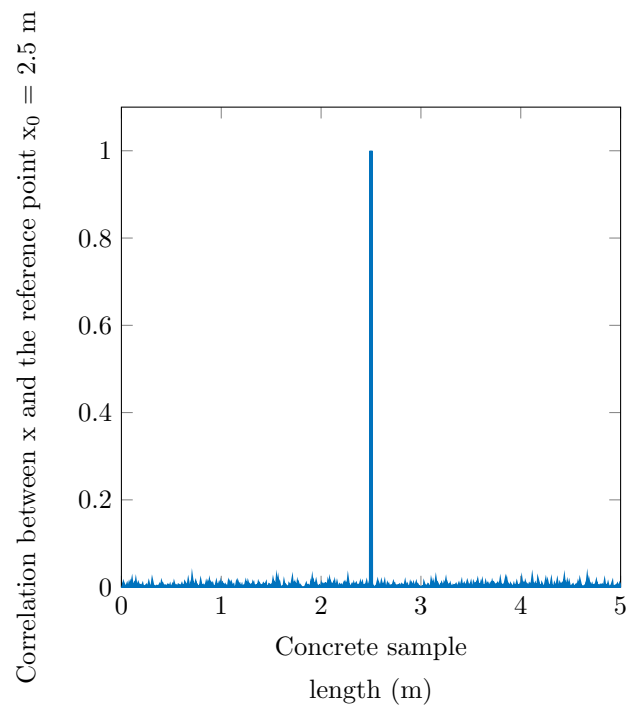


Figure 7: Evolution of the correlation over the beam with respect to the reference point $x_0 = 2.5$ m considering 100 000 realizations of the random field and a Gaussian distribution

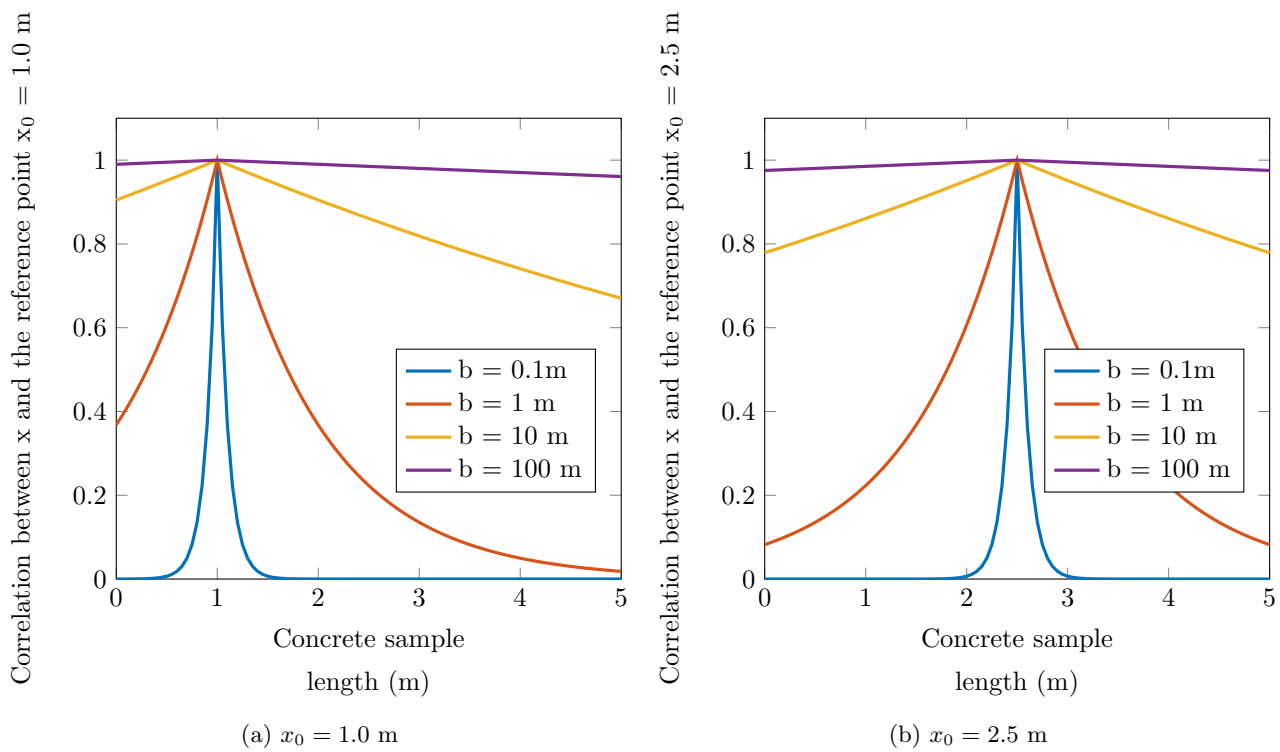


Figure 8: Evolution of the autocorrelation function of the one-dimensional random field over the beam for different values of the correlation length with respect to different reference points x_0

5 Two random variables

Remarks:

- One of the important properties of the multivariate normal density is that all conditional and marginal probabilities are also normal.
- $\rho_{E\nu} = 0.01$, $\Rightarrow E$ and ν are not independent.

13. Represent the joint probability density function.

Matlab file: `two_variables.m`

$\sigma_E = 5\% = 0.05 \times 21000\text{MPa} = 1050\text{MPa}$,

$\sigma_\nu = 10\% = 0.10 \times 0.3 = 0.03$

Bivariate Gaussian probability density function

```
pi=3.14;

sigmaE = 1050 ;
sigmaNu = 0.03 ;

muE = 21000 ;
muV = 0.3 ;

rho = 0.0 ;

E1 = 16000 ;
E2 = 26000 ;

nu1 = 0 ;
nu2 = 0.50 ;

nb_E = 50.0 ;
nb_nu = 50.0 ;

y0 = zeros(nb_E,nb_nu);

for i = 1:(nb_E+1)
    for j = 1:(nb_nu+1)
        E = E1 + (i-1)*(E2-E1)/nb_E ;
        nu = nu1 + (j-1)*(nu2-nu1)/nb_nu ;
        y0(i,j) = 1/(2*pi*sigmaE*sigmaNu*sqrt(1-rho^2))*exp(-1/(2*(1-rho^2))*((E-muE)^2/sigmaE^2
        + (nu-muV)^2/sigmaNu^2 -2*rho*(E-muE)*(nu-muV)/sigmaE/sigmaNu));
    end
end

E = (E1:(E2-E1)/nb_E:E2);
nu = (nu1:(nu2-nu1)/nb_nu:nu2);

surf(E,nu,y0)
```

```
set(gca,'XTick',[16000 18000 20000 22000 24000 26000],'YTick',[0 0.2 0.4],'ZTick',[0 0.002 0.004 0.006])
axis([16000 26000 0 0.5 0 0.006])

xlabel('E [MPa]','FontSize',25)
ylabel('\nu','FontSize',25)
zlabel('f_{E\nu}(E,\nu)','FontSize',25)

set(gca,'FontSize',20,'fontName','Times');
print(gcf,'-depsc2','TwoVariableGauss.eps');
```

- Does this function satisfy the condition:

$$\int_{\omega_X \omega_Y} f_{XY} dx dy = 1?$$

Matlab file: two_variables.m

```
fun = @(E,nu) 1 ./ (2 *pi*sigmaE.* sigmaNu*sqrt(1-rho^2)) .*
exp(-1/(2*(1-rho^2)) .* ( (E - muE).^2 ./ sigmaE.^2 + (nu - muV).^2 ./ sigmaNu.^2
-2*rho*(E-muE).*(nu-muV)/(sigmaE*sigmaNu))) ;
```

```
%p_Grenzen = quad2d(fun, 0, v_Grenze, 0, E_Grenze)
```

```
Emin = 0 ;
Emax = 300000 ;
```

```
numin = 0.0 ;
numax = 1.00 ;
```

```
q2 = integral2(fun,Emin,Emax,numin,numax);
or
q2 = quad2d(fun,Emin,Emax,numin,numax);
```

This integral equals to 1.

- Represent the marginal probability density function of ν . Estimate the probability that $\nu = 0.25$

The marginal probability density $f_\nu(\nu)$ is defined as:

$$\begin{aligned} f_\nu(\nu) &= \int_{\omega_E} f_{E\nu}(E, \nu) dE \\ &= \int_{-\infty}^{+\infty} \frac{1}{2\pi\sigma_E\sigma_\nu} \exp\left[-\frac{1}{2(1-\rho_{E\nu}^2)} \left[\frac{(E-\mu_E)^2}{\sigma_E^2} + \frac{(\nu-\mu_\nu)^2}{\sigma_\nu^2} - \frac{2\rho_{E\nu}(E-\mu_E)(\nu-\mu_\nu)}{\sigma_E\sigma_\nu} \right] \right] dE \\ &= \frac{\sqrt{2\pi}\sigma_\nu}{2\pi\sigma_E\sigma_\nu} \exp\left[-\frac{1}{2} \frac{(\nu-\mu_\nu)^2}{\sigma_\nu^2}\right] \\ f_\nu(\nu) &= \frac{1}{\sqrt{2\pi}\sigma_E} \exp\left[-\frac{(\nu-\mu_\nu)^2}{2\sigma_\nu^2}\right]. \end{aligned}$$

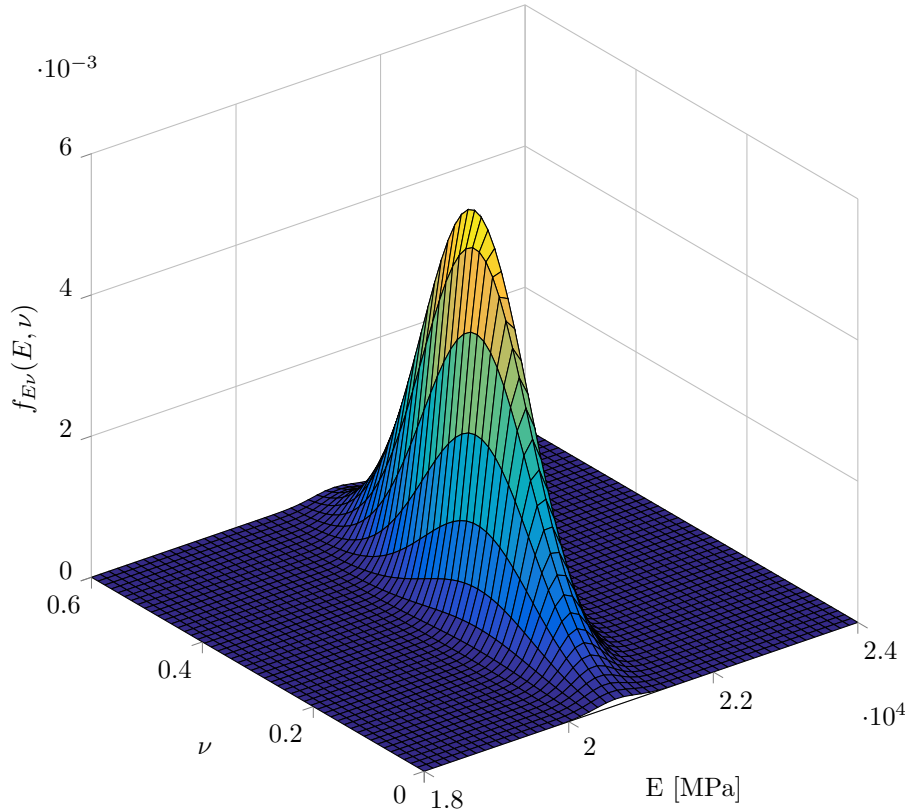


Figure 9: Joint probability density function $f_{E\nu}$

The probability that $\nu = 0.25$ is obtained through:

$$f_{\nu}(\nu = 0.25) = \frac{1}{\sqrt{2\pi} \times 1050} \int_{\nu=0.245}^{0.255} \exp -\frac{(\nu - 0.3)^2}{2 \times 0.03^2} d\nu,$$

to be estimated using Matlab as done previously.

15. Represent the marginal probability density function of E . Estimate the probability that $E = 21000$ MPa.

The marginal probability density $f_E(E)$ is defined as:

$$\begin{aligned} f_E(E) &= \int_{\omega_{\nu}} f_{E\nu}(E, \nu) dE \\ &= \int_0^{+\infty} \frac{1}{2\pi\sigma_E\sigma_{\nu}\sqrt{1-\rho_{E\nu}^2}} \exp -\frac{1}{2(1-\rho_{E\nu}^2)} \left[\frac{(E-\mu_E)^2}{\sigma_E^2} + \frac{(\nu-\mu_{\nu})^2}{\sigma_{\nu}^2} - \frac{2\rho_{E\nu}(E-\mu_E)(\nu-\mu_{\nu})}{\sigma_E\sigma_{\nu}} \right] d\nu \\ &= \frac{\sqrt{2\pi}\sigma_E}{2\pi\sigma_E\sigma_{\nu}} \exp -\frac{1}{2} \frac{(E-\mu_E)^2}{\sigma_E^2} \\ f_E(E) &= \frac{1}{\sqrt{2\pi}\sigma_{\nu}} \exp -\frac{(E-\mu_E)^2}{2\sigma_E^2} \end{aligned}$$

The probability that $E = 21000$ MPa is obtained through:

$$f_E(E = 21000\text{MPa}) = \frac{1}{\sqrt{2\pi} \times 0.03} \times \int_{E=20900\text{MPa}}^{21100\text{MPa}} \exp -\frac{(E - 21000)^2}{2 \times 1050^2} dE,$$

to be estimated using Matlab as done previously.

16. Estimate the probability that $E = 21000$ MPa while $\nu = 0.25$

$$\begin{aligned}
 A_4 &= \mathcal{P}\{E = 21000\text{MPa}, \nu = 0.25\} \\
 &= \mathcal{P}\{20900\text{MPa} < E < 21100\text{MPa}, 0.245 < \nu < 0.255\} \\
 &= \int_{\mathcal{B}_E} \int_{\mathcal{B}_\nu} f_{E\nu}(E, \nu) dE d\nu \\
 &= \int_{E_1=20900\text{MPa}}^{E_2=21100\text{MPa}} \int_{\nu_1=0.245}^{\nu_2=0.255} \frac{1}{2 \times \pi \times 1050 \times 0.03} \exp\left[-\frac{(E - 21000\text{MPa})^2}{2 \times 1050^2\text{MPa}^2} - \frac{(\nu - 0.3)^2}{2 \times 0.03^2}\right] dE d\nu \\
 &= 0.00505 \int_{E_1=20900\text{MPa}}^{E_2=21100\text{MPa}} \int_{\nu_1=0.245}^{\nu_2=0.255} \exp\left[-\frac{(E - 21000\text{MPa})^2}{2 \times 1050^2\text{MPa}^2} - \frac{(\nu - 0.3)^2}{2 \times 0.03^2}\right] dE d\nu \\
 A_4 &= 0.0025365
 \end{aligned}$$

Matlab file: two_variables.m

```

fun = @(E,nu) 1 ./ (2 .* pi() .* sigmaE .* sigmaNu*sqrt(1-rho^2))
.* exp(-0.5 .* ( (E - muE).^2 ./ sigmaE.^2 + (nu - muV).^2 ./ sigmaNu.^2 )) ;

Emin = 20900 ;
Emax = 21100 ;
numin = 0.245 ;
numax = 0.255 ;
% q1 = quad2d(fun, v_min, v_max, E_min, E_max) To see which is the best estimation
q2 = integral2(fun,Emin,Emax,numin,numax);

```

17. Estimate the probability that $E < 21000$ MPa while $\nu < 0.25$

$$\begin{aligned}
 A_5 &= \mathcal{P}\{E < 21000\text{MPa}, \nu < 0.25\} \\
 &= \int_{\mathcal{B}_E} \int_{\mathcal{B}_\nu} f_{E\nu}(E, \nu) dE d\nu \\
 &= \int_{E_1=0\text{MPa}}^{E_2=21100\text{MPa}} \int_{\nu_1=0.0}^{\nu_2=0.255} \frac{1}{2 \times \pi \times 1050 \times 0.03} \exp\left[-\frac{(E - 21000\text{MPa})^2}{2 \times 1050^2\text{MPa}^2} - \frac{(\nu - 0.3)^2}{2 \times 0.03^2}\right] dE d\nu \\
 &= 0.00505 \int_{E_1=0\text{MPa}}^{E_2=21100\text{MPa}} \int_{\nu_1=0.0}^{\nu_2=0.255} \exp\left[-\frac{(E - 21000\text{MPa})^2}{2 \times 1050^2\text{MPa}^2} - \frac{(\nu - 0.3)^2}{2 \times 0.03^2}\right] dE d\nu \\
 A_5 &= 0.25
 \end{aligned}$$

18. Are the two random variables E and ν independent?

The two random variables are not independent because $\rho_{E\nu} = 0.01$:

- Independent $\Rightarrow \rho = 0$ so, $\rho \neq 0 \Rightarrow$ not independent.

but, be careful: $\rho = 0 \nRightarrow$ Independent. See for Example, Exercise 2.

19. Plot the conditional probability of E , given $\nu = \bar{\nu}$ for $\rho = 0.0$ and for $\rho = 0.4$.

The conditional probability is defined as:

$$f_{E|\nu}(E) = \frac{f_{E\nu}(E, \nu)}{f_\nu(\nu)} \quad \text{for } f_\nu(\nu) > 0.$$

$$f_{E|\nu}(E) = \frac{\frac{1}{2\pi\sigma_E\sigma_\nu\sqrt{1-\rho_{E\nu}^2}} \exp\left[-\frac{1}{2(1-\rho_{E\nu}^2)} \left[\frac{(E-\mu_E)^2}{\sigma_E^2} + \frac{(\nu-\mu_\nu)^2}{\sigma_\nu^2} - \frac{2\rho_{E\nu}(E-\mu_E)(\nu-\mu_\nu)}{\sigma_E\sigma_\nu} \right] \right]}{\frac{1}{\sqrt{2\pi}\sigma_E} \exp\left[-\frac{(\nu-\mu_\nu)^2}{2\sigma_\nu^2}\right]} \quad (9)$$

$$f_E(E) = \frac{1}{\sqrt{2\pi}\sigma_\nu} \exp\left[-\frac{(E-\mu_E)^2}{2\sigma_E^2}\right] \quad (10)$$

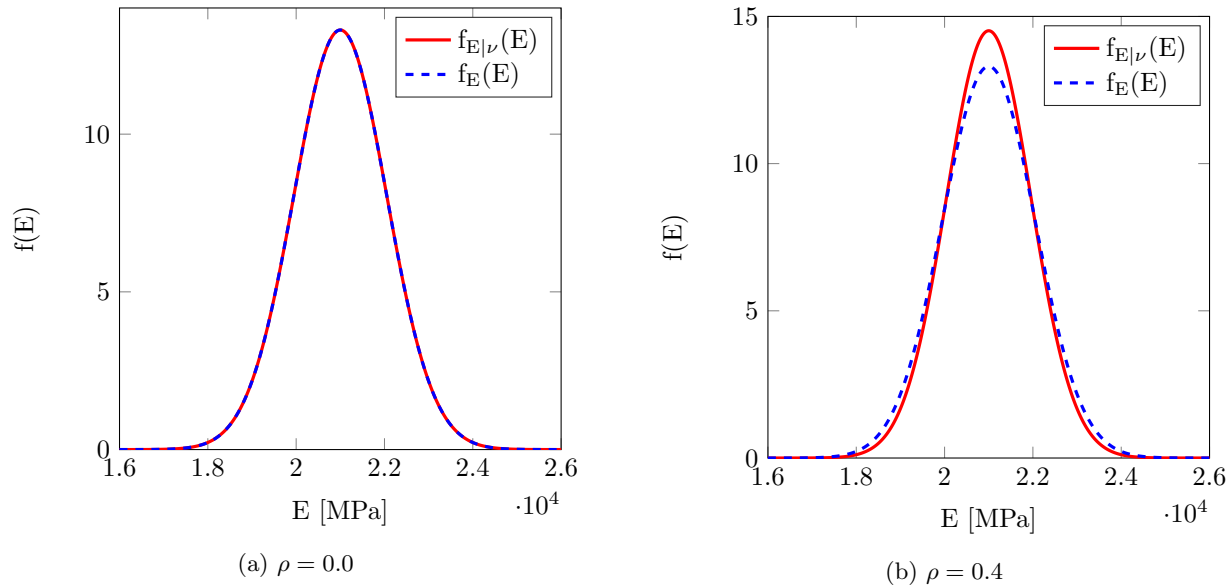


Figure 10: Marginal probability density function and conditional probability density function for the Young modulus

Matlab file: two_variables.m

```
pi = 3.14;
sigmaE = 1050 ; sigmaNu = 0.03 ;
muE = 21000 ; muV = 0.3 ;
rho = 0.40 ; nu = 0.30 ;
E1 = 16000 ; E2 = 26000 ;
nb_E = 50.0 ;

for i = 1:(nb_E+1)
E = E1 + (i-1)*(E2-E1)/nb_E ;
y0(i) = 1/(sqrt(2*pi)*sigmaNu)*exp(-(E-muE)^2/(2*sigmaE^2));
y01(i) = 1/(2*pi*sigmaE*sigmaNu*sqrt(1-rho^2))*exp(-1/(2*(1-rho^2))*((E-muE)^2/sigmaE^2 + (nu-muV)^2...
/sigmaNu^2 - 2*rho*(E-muE)*(nu-muV)/sigmaE/sigmaNu))/(1/(sqrt(2*pi)*sigmaE)*exp(-(nu-muV)^2/(2*sigmaNu^2)))
end

E = (E1:(E2-E1)/nb_E:E2);
```

```
plot(E,y01,'r');  
hold on;  
plot(E,y0,'b')  
  
xlabel('E [MPa]', 'FontSize', 25)  
ylabel('f(E)', 'FontSize', 25)  
  
legend('f_{E|\nu}(E)', 'f_{E}(E)')  
legend('boxoff')  
  
set(gca, 'FontSize', 25, 'fontName', 'Times');  
print(gcf, '-depsc2', 'ConditDensFunctions.eps');
```